Network Models, Data Link and IP Addressing

Computer Networks - Units 1-2 - Foundations

Why layers?

Layering splits the complex job of communication into smaller services. Each layer talks to the layer above and below through interfaces and to its peer on the other host through a protocol.

OSI vs TCP/IP

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OSI layer        TCP/IP layer    Examples
7 Application    Application     HTTP, DNS, SMTP
6 Presentation   Application     TLS, encoding
5 Session        Application     sessions
4 Transport      Transport       TCP, UDP
3 Network        Internet        IP, ICMP
2 Data link      Link            Ethernet, Wi-Fi
1 Physical       Link            cables, radio
```

Data link layer

- Framing, MAC addressing (48-bit), error detection with CRC.
- Flow control: stop-and-wait, sliding window (Go-Back-N, Selective Repeat).
- Switches forward frames using a MAC address table learned from source addresses.

IPv4 addressing and subnetting

An IPv4 address has 32 bits written as four decimal octets. CIDR notation a.b.c.d/n means the first n bits identify the network.

> *Example: 192.168.10.0/26 -> 6 host bits, 2^6 = 64 addresses, 62 usable hosts (minus network and broadcast). Subnet mask 255.255.255.192; the next subnet starts at 192.168.10.64.*

- Private ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- NAT lets many private hosts share one public address.
- IPv6 uses 128-bit addresses to solve address exhaustion.